Final Project Report

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Specification Table

Base Model	Repeated Prisoners Dilemma
Extension assumptions	Repeated Prisoners Dilemma extended onto a game with multiple people. The multiple people will not all be able to be rational and will sometimes act randomly. The game will punish people if a lot of people take the betray option or reward them if there are a few.
Techniques showcased	1.Monte Carlo simulations 2.game theory
Modelling question 1	Using Monte Carlo simulations to simulate irrational decisions made by the prisoners to see how common strategies work against non-consistent prisoners?
Modelling question 2	Is there an optimal strategy against a group with irrational prisoners? Does more prisoners or varying abilities of the prisoner affect the optimal strategy.

Introduction

The regular prisoner's dilemma is a two-player game. However, many games or situations can have more than one player. One such game is Fortnite which is about gathering loot and fighting a bunch of other players in a free for all. In such a game, if everyone lands in one location then everyone gets less loot and if only one person visits that location then that player gets a large amount of loot, so they will have a greater advantage beating the other players. There is also an option to visit a safer place with less loot. This resembles a prisoner's dilemma question but with more than one opponent. I would like to analyse if there is an optimal strategy in a situation like this. Additionally, in games like Fortnite, your opponents are randomly selected and thus will not always be all competent and follow strategy. These opponents will make irrational decisions based on random chance or gut feeling. It would be interesting to see how common strategies for the Repeated prisoners Dilemma hold up against some random opponent and to see if there is a potentially better strategy. This can be simulated with a monte carlo simulation by trying the strategies against a bunch of randomly generated different opponents. Consider if there is an optimal strategy against a group of random people and is it different to the normal solutions. This can be solved with use of game theory to quantify and check if there are other strategies that work. We can combine all

the other players decisions into one person. This means we can just solve it with a two-person game theory solver.

Model Description

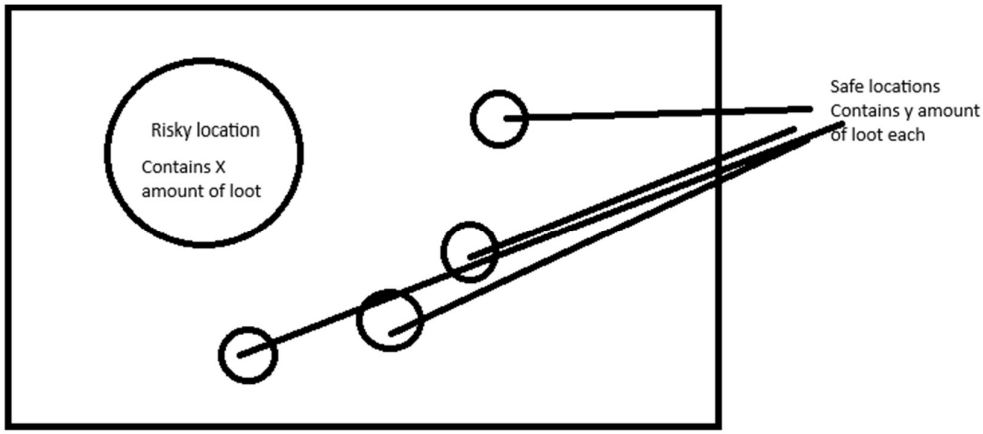

Figure 1: Model visualization

The regular prisoner's dilemma is when two people have a choice between betraying or allying with the other person. If only one of them betray then the betrayer is rewarded a lot and the ally loses, if both people betray then they both lose a lot. Finally, if both people ally then nothing is lost. To expand on this model, we need to consider how a multiple people Prisoner's dilemma will work. Since this model is inspired by Fortnite, we can choose to go to two locations. A safe location where the loot is guaranteed to be a certain value and a risky location where the loot is shared between all the people that go there. This makes is like Fortnite as there are many safe locations to travel to but have considerably less loot. We can denote the loot in the safe location as y and the loot in the risky location as x. We can also denote the people that goes to the safe location as n_2 and n_1 for the people that go to the risky location. If all the player goes to the risky location, then the loot is split evenly among all the people. Since there is $n_1 + 1$ people there then the point gained is $\frac{x}{n_1+1}$ for the player and $\frac{x\,n_1}{n_1+1}$ for the other players. If the player goes to the safe location, then all the loot in the risky location is given to the other players and the player gets y loot. In this model there we assume that no players will die or lose before each loot run as to keep the amount of people the same every iteration. This is because we can assume the players are not fighting yet and still gathering loot before the battle. There is also the assumption that the loot is shared evenly between the players. This makes the model simpler as we do not need to consider a changing amount of people.

The actions of the other group will not always be completely random. They can have some preference in the risky location or the safe location. This can be modelled by

changing n1 and n2 to s1, s2 and r1, r2 for the group that prefers the safe location and the group that prefers the risky location respectively. For the rest of the players, we add up all their scores to calculate it more efficiently. Therefore, for each run we add the total loot gathered by every other player from both locations. To compare with the results, we can divide it by number of other players. This gives us our score against the average of their scores. This gives us a decent idea on how we would fare against the other players with decent accuracy as the outliers won't be too big and will fall between reasonable bounds.

Risky location loot = x, Safe location loot = y

r = r1 + r2, s = s1 + s2

For one loot run:

		Other Players combined	
		Prefer Risky Location	**Prefer Safe Location**
Main Player choice	*Risky Location*	$\frac{x}{r_1+1}, (\frac{x\, r_1}{r_1+1} + r_2 \times y)/r$	$\frac{x}{s_1+1}, (\frac{x\, s_1}{s_1+1} + s_2 \times y)/s$
	Safe Location	$y, (x + r_2 \times y)/r$	$y, (x + s_2 \times y)/s$

Normally a game will not end in just one loot run. This will happen multiple times during one game, so the goal is to gather the most loot as possible. If you have more loot than someone then you are more likely to win. To compare how well the player is doing you can find the average loot obtained from every other player by dividing and seeing if the main player has more than the average. This lets you discover if you have the advantage against the average player. The common strategies we are looking into for the players are always risky, always safe, randomly pick and to follow the location that gives each player the most points.

This model is obviously discrete as it occurs n number of times with each time a discrete state. This results in many non-linear states as the increase in loot is not always linear. Since there is random chance based on the decision of each person, the model is stochastic and not deterministic. This results in a numerical analysis as we are trying to approximate the results.

A Markov chain can be used to further model the decision-making process of the other players and players during the game. We can have the four states be where each state contains if they prefer the risky location or safe location for the main player and if they prefer risky location or safe location for the other players. This can be used to have a

more sophisticated decision making by allowing players to react to the decisions of the previous runs.

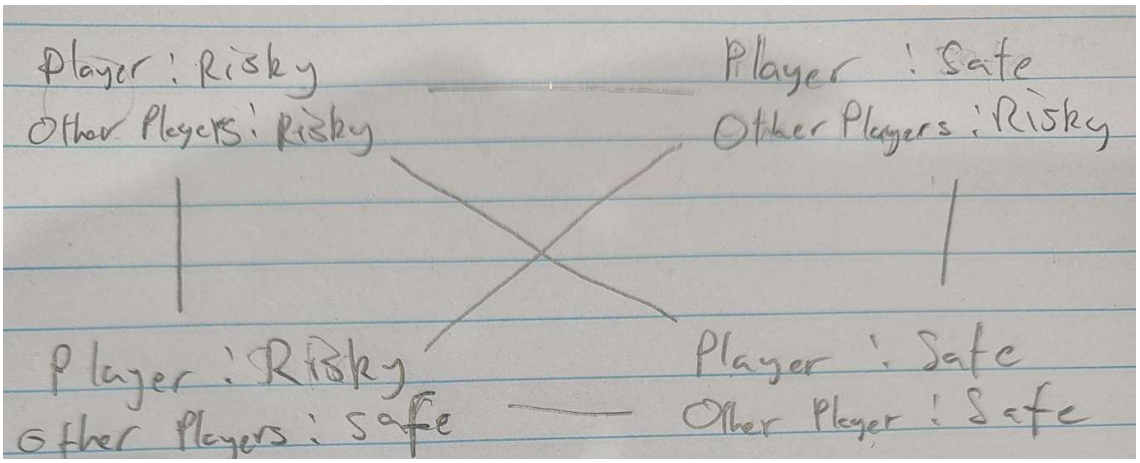

Figure 2 Markov Chain for decision of Players

Results

First, we can check what happens to the common strategies when there is an equal chance for the other players to go to the risky location and the safe location. We can simulate this with a Monte Carlo simulation with a random number generator to pick which place the other players go.

The starting values is 99 for the number of other players. This is because in a normal game of Fortnite there are 100 players with you being one of them. The amount of loot in the risky location will be 50 while the amount of loot in the safe locations will be 1 because setting the risky place should punish if a decent amount of people goes to that location while rewarding people if there is less than half the people go there. We shall do 1000 repeats to get a good enough approximation. There will be 3 loot runs during each repeat as the average Fortnite game is not that long and going through 3 locations will give you enough loot to take people on.

Opponent player has 50% chance for each location:

	Average Player Score	**Average Opponent Score**
Always Safe	3.0	3.0
Always Risky	3.0	3.0
Random Choice	3.0	3.0
Follow Most Loot	3.0	3.0

Results of the first 8 runs:

Run number	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Always Safe	3.0,3.0	3.0,3.2	3.0,3.1	3.0,3.1	3.0,3.1	3.0,3.1	3.0,2.9	3.0,3.1

Always Risky	2.9,2.9	3.1,3.0	2.8,2.9	3.0,3.0	3.1,3.0	3.0,3.0	3.2,3.1	3.0,3.0
Random Choice	3.0,3.1	2.7,2.8	2.9,3.0	2.8,2.9	3.0,3.0	3.0,3.1	2.9,2.9	3.0,3.0
Follow Most Loot	2.9,3.0	3.1,3.0	2.9,3.0	2.9,3.0	3.1,3.1	3.0,3.0	3.3,3.1	2.9,3.0

The results show that in the long run, all the choices produce the same average player and average opponent score. There seems to be no effect in the choice of strategy has on the score you get if the other players have a 50% chance to pick each location.

The always safe option has a consistent result of 3 loot. This however does not beat the average of the other players most of the time as shown with the first 8 runs. Always risky seems to fluctuate a bit more, while random choice seems to have the most possible variation. These will be averaged out with a huge number of games. The strategies seem to all have different pros and cons where with the follow most loot seemingly the highest risk but the highest rewards. In a game of Fortnite if you have better equipment doesn't mean you will instantly win but will have a better chance of winning. A strategy like follow most loot will get you games with a lot more extra loot on some games which makes those games a lot easier to win while only being behind a tiny amount of loot normally which is only slightly harder to win. It depends on how much you value easy games or how much you value consistency for the best decision.

Prefer Risky Location:

	Average Player Score	**Average Opponent Score**
Always Safe	3.0	2.3
Always Risky	2.0	2.2
Random Choice	2.5	2.3
Follow Most Loot	3.0	2.3

Results of the first 8 runs:

Run number	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Always Safe	3.0,2.1	3.0,2.2	3.0,2.2	3.0,2.2	3.0,2.2	3.0,2.3	3.0,2.3	3.0,2.2
Always Risky	2.1,2.3	2.0,2.2	2.0,2.3	1.9,2.2	2.1,2.3	1.9,2.2	2.1,2.1	1.9,2.2
Random Choice	2.4,2.2	3.0,2.3	2.6,2.2	2.7,2.3	2.6,2.3	2.7,2.3	2.7,2.2	2.7,2.2
Follow Most Loot	3.0,2.3	3.0,2.2	3.0,2.3	3.0,2.3	3.0,2.3	3.0,2.3	3.0,2.4	3.0,2.4

The results are obtained by giving the players a 75% chance to go to the risky location. By comparing the results, it shows that the best choices are follow most loot and the always safe option. Always risky is never a good choice as a lot of people going to the risky area will reduce the loot for everyone. By picking the random option will be better then just picking the risky location as randomly you will pick the safe location and earn more loot. The best strategy is the to always play safe and to follow the most loot for the average runs. This is because both methods will pick the safe option as the safe option

is always going to generate more total loot. This is due to there is little loot for everyone that goes to the risky location because everyone is going there. The safe location will give more loot so follow most loot will always go to the safe location. Always risky is a terrible choice as you will always pick the risky option which is the location with the least amount of loot. In the sample runs, the always risky loses all the time. This can be attributed to the fact that even though most people pick the risky option most of the time, there still is a significant amount of people that pick the safe location. These people will earn decent points so always risky is a very terrible strategy in this situation. The random choice is better than the opponent on average but the gap is small so there is a still a small chance that random choice could lose. Looking at run 1 for the sizeable choices, there is only a difference of 0.2. Always safe and Follow most loot always has a sizeable difference between the choices as they produce a consistent 3 score while the other players float at around 2.2 to 2.3. Additionally, these strategies result in the exact same amount of loot. This can benefit by having each run being roughly the same and not deal with runs with less loot or runs with more loot. This gives the players a good understanding of what each run is expected to be like and gives them a good idea of planning out how to use the loot from the start.

Prefer Safe Location:

	Average Player Score	**Average Opponent Score**
Always Safe	3.0	3.8
Always Risky	6.0	3.7
Random Choice	4.5	3.7
Follow Most Loot	5.0	3.7

Results of the first 8 runs:

Run number	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Always Safe	3.0,3.7	3.0,3.7	3.0,3.8	3.0,3.8	3.0,3.9	3.0,3.8	3.0,3.8	3.0,3.7
Always Risky	5.6,3.7	5.4,3.6	6.1,3.7	5.9,3.7	8.6,3.8	6.4,3.7	5.9,3.7	5.9,3.7
Random Choice	3.7,3.7	3.9,3.8	3.0,3.8	4.5,3.7	4.5,3.6	3.0,3.8	4.8,3.7	5.5,3.7
Follow Most Loot	4.8,3.7	4.4,3.6	4.4,3.7	7.2,3.9	5.5,3.7	4.7,3.7	5.4,3.7	4.6,3.7

These results are obtained by giving the opponent player a 75% chance to go to the safe location. The best strategy during this is to pick always risky. This seems to be because that few players are picking risky then most of the loot in risky is not looted so we can always go there for the most loot. Always safe is by far the worst strategy as the risky place will always have more loot. There is no incentive to go to the safe location. Random choice also does well as there is a 50% chance to pick either option so it will occasionally pick the risky option with more loot. This does beat the average opponent

score most of the time. However as shown in run number 1, 3 and 6 for random choice results in that choice drawing or losing. This shows that it is not always a consistent option and is a high-risk high reward option. Follow most loot looks more consistent than random choice and results in it having a higher a 0.5 higher than random choice. This is because it will pick safe location then switch to risky location due to the risky location doing well. However, this limits the player from going to the risky option on the first try unlike random chance where there is a small chance that it only picks the risky location. This will make the option weaker than always Risky. For the options always risky, random choice and follow most loot will have a huge difference in scores from one run to another even with the same actions. This can be observed with always risky run 2 having 5.4 and run 5 having 8.6. This is very different from when opponent has 50% player and prefer risky location. This indicates that loot becomes volatile if there are less people that go to the risky location. It could be because the math is changed with less people going to the risky location the loot given is increased. Consider if only 5 people goes to the risky location, then each will receive 10 and for 4 people there will be 12.5 each. This has a difference of 2.5 loot. Now if 50 people go to the risky location there is only 1 loot each and 49 people going then there is 1.02 loot each. This is a far smaller increase of 0.02. Since if there are less people then it will increase and decrease far more with a little change explaining why it is so volatile.

Comparing the four strategies in all three situations. Both choose only risky and only safe only benefit more on one of prefer safe location and prefer risky location respectively. The best option seems to be follow most loot and then random. They both on average provide more loot for all situations. Provide more loot does give more on average than random but it is hard for a player to predict how much loot that is earned during each loot run. If you go to the safe location, you will have no clue what happens in the risky location as the loot you get is not dependent on the other people that went there. It would require a method to figure out what your opponents are doing for this strategy to be achievable. The best realistic strategy is to pick random as it gives the player an edge no matter what the ratio of people going to each location is. This is if you don't know the preferences of the other players. However, the highest score for the enemy player is to take the safe location. It could be that players would prefer the safe location so potentially always going risky could work. The safest method would be to follow the most loot but is unrealistic. The best method is to be completely random.

Let us check if there are any mix strategies for this model. We can also consider the other players action by allowing them to pick where they will be distributed. By having prefer risky to be 75% people that will go to the risky and prefer safe to be the 75% players that prefer safe. We can create the table below.

SINGLE ROUND:	PREFER RISKY	PREFER SAFE
RISKY LOCATION	0.7,0.8	1.9, 1.2
SAFE LOCATION	1, 0.8	1, 1.3

The Nash equilibrium is to always for the player to pick risky location and for the other players to favour playing safe. This follows the conclusion obtained previously as by picking prefer always safe, the average opponent score is by far the highest. This results in a player advantage as the loot obtained by the player is far greater. Picking the safe location has the greatest increases in points. However, if the player is always trying to go to the risky location, the other players can decide to prefer doing the risky action which makes it so they have a slight advantage. This however is not very consistent as the advantage is very slight. For a substantial lead, the player must pick the safe location, and the other players also must join in for the safe location. Picking the safe location just doesn't give enough rewards then to always go for the risky location. Although picking the safe location for the other players will reward the players with the most loot. However, it is also optimal for the other players to take the risky location as the difference between the average player is very small. This means that the net advantage from the players is tiny which is preferable for being stronger with more loot against stronger enemies.

SINGLE ROUND:	PREFER RISKY	PREFER SAFE
RISKY LOCATION	0.8,0.9	1.2, 1.1
SAFE LOCATION	1, 0.9	1, 1.1

The prefer risky and prefer safe was set to 60%. This is to see if the strategy changes if it is closer to completely random. This results in the same strategy as before for the Nash equilibrium. However, all the choices result in players and other players to have the same loot. The best strategy for this is depending on the preference if the players like having higher amounts of loot or lower amounts of loot.

SINGLE ROUND:	PREFER RISKY	PREFER SAFE
RISKY LOCATION	0.6,0.6	4.5, 1.4
SAFE LOCATION	1,0.6	1, 1.4

The prefer risky and prefer safe was set to 90%. The Nash equilibrium is the same as the other 2. This makes sense as it is the greatest reward for both the player and the other players. For the other player to choose to prefer risky then they would be heavily disadvantaged as they always lose against the player or come out even. Picking to prefer safe would give them a higher chance to win if the player picks the safe location. This however is balanced by a huge chance to lose if the player picks risky. The loot obtained is 4.5 for the player and only 1.4 for the other players. The player would already

pick the risky location as it guarantees them a benefit so the best option for the other players would also be to pick the risky location.

There is very little incentive for player to pick anything else but to go to the risky location. Additionally, the best option for the other players is to also pick the risky location which contradicts the Nash equilibrium. This is because the difference in loot between the players and other players gets too big if they pick to prefer the safe location. It is always better to prefer the risky location as the difference between the score is smaller.

We can use the results from here to give an estimate of the odds for the transition matrix of the Markov chains. If the player and other player is at the risky location, they would favour staying at the risky location as switching to another location for both players would result in a not favourable position for both players. If the player is at safe and the other player is at the risky location, then player would try to move to risky while the other player will try to move to safe. When the player is at risky and the other player is at safe then the player would have no incentive to switch to safe. There is an incentive for the other players to switch to risky as an attempt to gain rewards but not as big as the players incentive to switch.

Player | Other Player

Row 1 is Risky | Risky, Row 2 is Safe | Risky, Row 3 is Risky | Safe, Row 4 is Safe | Safe

Column 1 is Risky | Risky, Column 2 is Safe | Risky, Column 3 is Risky | Safe, Column 4 is Safe | Safe

$$P = \begin{pmatrix} 0.6 & 0.1 & 0.1 & 0.2 \\ 0.3 & 0.1 & 0.3 & 0.3 \\ 0.3 & 0.1 & 0.5 & 0.1 \\ 0.3 & 0.2 & 0.4 & 0.1 \end{pmatrix}$$

$$P^3 = \begin{pmatrix} 0.4 & 0.1 & 0.3 & 0.2 \\ 0.4 & 0.1 & 0.3 & 0.2 \\ 0.4 & 0.1 & 0.3 & 0.2 \\ 0.4 & 0.1 & 0.3 & 0.2 \end{pmatrix}$$

$$P^{10} = \begin{pmatrix} 0.4 & 0.1 & 0.3 & 0.2 \\ 0.4 & 0.1 & 0.3 & 0.2 \\ 0.4 & 0.1 & 0.3 & 0.2 \\ 0.4 & 0.1 & 0.3 & 0.2 \end{pmatrix}$$

By multiplying the matrix with itself the matrix approaches P^{10}. This shows that it doesn't matter what the starting preference is, all the preferences have the same chance to end in each location. There is zero absorption as there is always a chance for each outcome to occur as the whim of players can be random. Since every state can be obtained from another state then the matrix is Ergodic. The matrix shows that the highest chance to end up on is Risky | Risky which makes sense as it is the safest option to choose. It results in the lowest difference between all players but has a slight benefit

to the main player. The second most common decision is for the player to go to the risky location while the other players go to the safe location. This is the same as the Nash equilibrium as it rewards both players with the most points. It however is not ideal to visit as the point difference between the players is too large so most players will attempt to go to the risky location. Since the player is almost always going to go to the risky location, then Safe | Risky and Safe | Safe are very unlikely to occur. There is little benefit to go there unless they expect that the other players will go to the risky location.

We can see what the results of the games through a Monte Carlo simulation. I have set the number of iterations to 1000 with 75% for each preference. The game will have 3 loot runs.

	Average Loot	**Has advantage**
Player	3.5	653
Other Player	3.0	347

Simply being able to pick your decisions instead of being apart of the group gives you a decent increase of loot. There is a substantial increase in the number of games where you have the advantage if you can strategize and plan to get loot.

The best option seems to be picking randomly. The first Monte Carlo simulations showed that picking random decisions is the best if you didn't know how many people went to a location. Just by picking randomly gives you an advantage if there is any bias to either location for the group. After looking at the payout for the choices of the players it seems like picking just risky is the best choice for the player as it denies the opponent from getting loots. By picking risky, the other players must mostly pick risky as well to get a large amount of loot. It seems like the best option is to pick randomly while favouring picking risky. This seems to be the best option.

List of algorithms and concepts

Game theory for the basis of the model

Monte Carlo to simulate random peoples' behaviours

Game theory for finding optimal strategy for all players

Markov chains to model behaviour

Monte Carlo to simulate a game with Markov chains

AI statement

None used